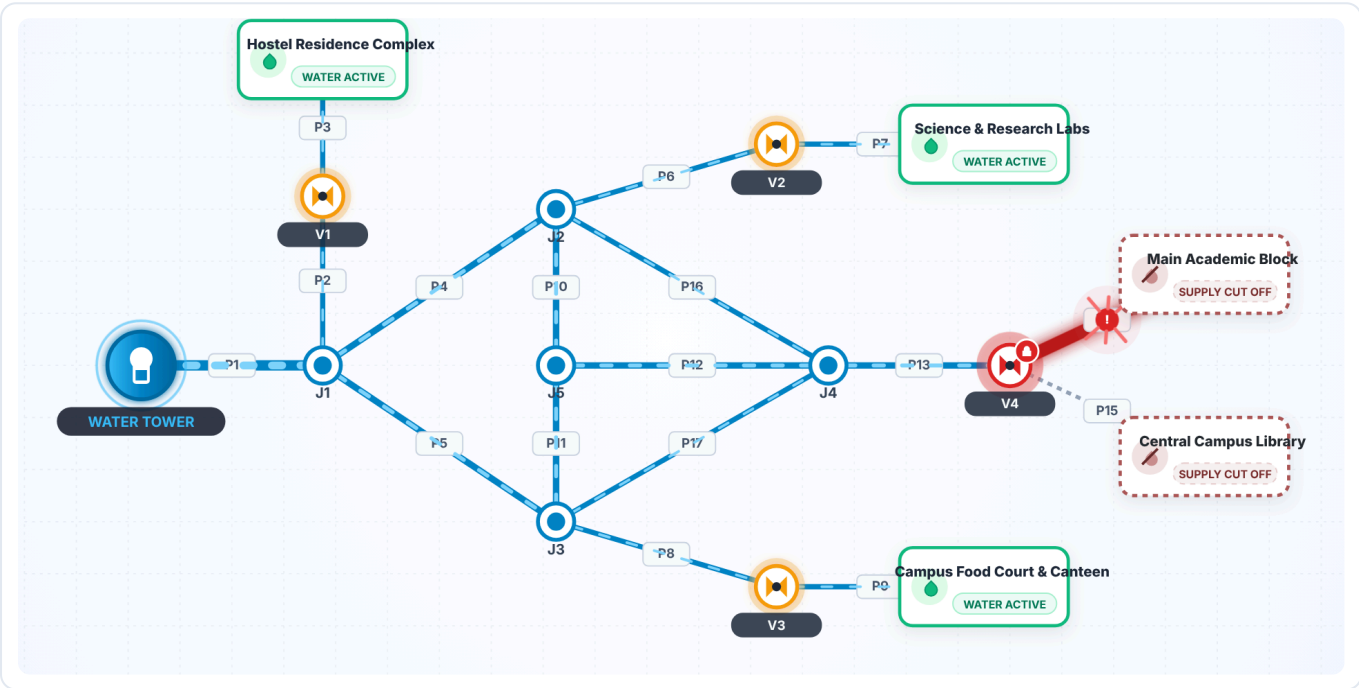
SIMPLE GUIDE · FUNDAMENTALS OF AI MICRO-PROJECT

Campus Water Network: Pipe-Burst Isolation with BFS

What the website does, how it works step by step, how the code is organised, and how to put it online with Render.



The website after a burst on pipe P14: valve V4 is closed and two buildings are cut off.

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Live site: kits-campus-water-bfs.netlify.app
Code: github.com/KrisJeffordImmanuvel/FAI_Sujith

1 What does this website do?

Imagine the water system of our campus: one **water tower**, a set of **pipes**, some **valves** that can be shut, and five **buildings** that need water.

Now imagine one pipe bursts. Two questions need answers quickly:

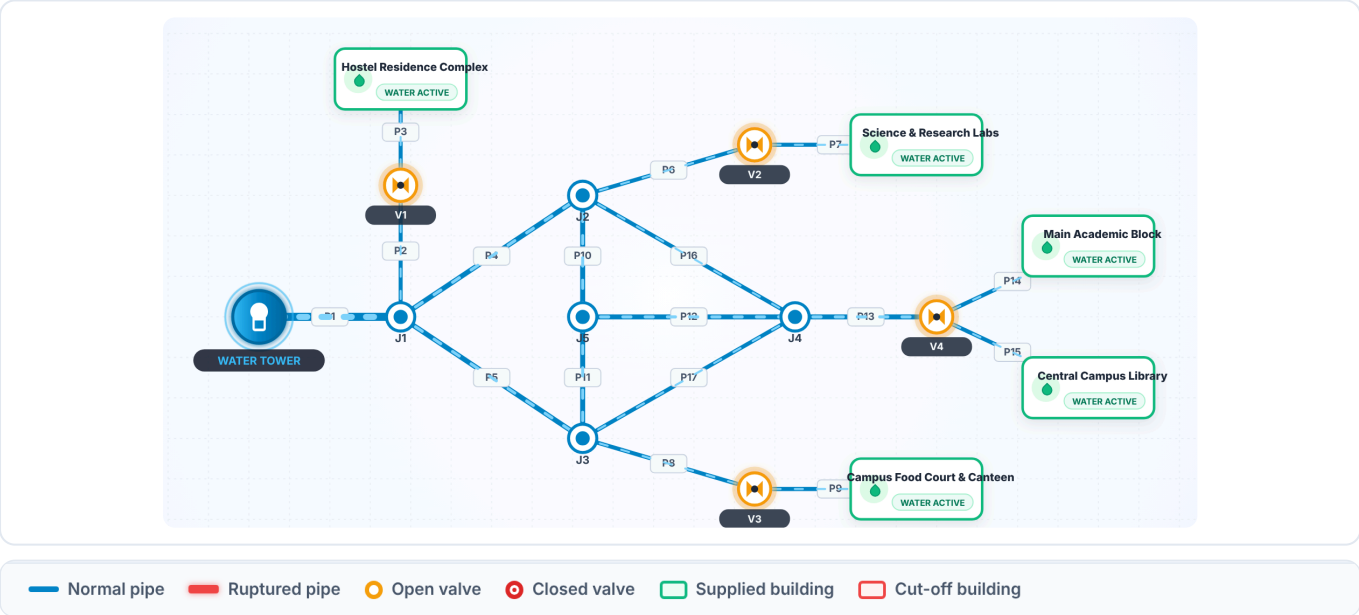
1. **Which valves should we close** to stop the leak?
2. **Which buildings will lose water** once those valves are closed?

The website answers both questions automatically using an algorithm called **Breadth-First Search (BFS)**, and it animates every step so you can see exactly how the answer was found.

> **In one sentence**
>
> You click a pipe to "break" it; BFS finds the nearest valves to close, then checks which buildings can still be reached from the water tower, and the site shows the result on a map and in a printable report.

2 The campus network

To a computer, the water system is a **graph**: a set of dots (called *nodes*) joined by lines (called *edges*). Every pipe is an edge; everything else is a node.



The network when nothing is broken, and the legend shown under the map on the website.

Part	How many	What it is
Water tower (WT)	1	Where all the water comes from.
Junctions (J1–J5)	5	Points where pipes meet and split.
Valves (V1–V4)	4	Gates that can be shut. There is one in front of each building group.
Buildings	5	Hostel, Research Labs, Food Court & Canteen, Academic Block, Library.
Pipes (P1–P17)	17	The connections. Some form *loops*, so there can be more than one route between two points.

3 What is BFS? (the simple version)

Think of dropping a stone into a pond. The ripple spreads out in rings: first the nearest water, then the next ring, then the next. **BFS explores a graph the same way**: first everything one pipe away, then everything two pipes away, and so on.

Ring 1
Nodes 1 pipe away from the start

Ring 2
Nodes 2 pipes away

Ring 3 ...
Keeps going until there is nothing new to visit

To keep the rings in order, BFS uses a **queue**. It works like the line at the canteen counter: whoever joins first is served first ("first in, first out"). BFS also keeps a **visited list** so it never checks the same node twice, even when pipes form a loop.

> **The key idea**
>
> BFS counts **hops** (how many pipes you cross), not metres. So when BFS spreads out from a burst pipe and stops at the first valve it meets on each side, it has found **the nearest valves that isolate the rupture (fewest hops)**.

The website uses BFS **twice**:

- **1st BFS:** starts at the burst pipe and spreads out until it hits valves. Those are the valves to close.
- **2nd BFS:** starts at the water tower and spreads through the open pipes. Any building it reaches still has water; any building it cannot reach is cut off.

4 How it works: the 5 steps

1. **Select pipe** — mark the burst
2. **Find boundary valves** — 1st BFS
3. **Close valves** — shut them
4. **Check supply** — 2nd BFS
5. **Report** — save as PDF

1 Select pipe

Click a pipe on the map, or choose it from the list. This is the pipe that “burst”.

You see: the pipe turns red with a burst symbol.

2 Find boundary valves (1st BFS)

BFS starts from *both ends* of the burst pipe at the same time and spreads outwards one pipe at a time. It never crosses the burst pipe itself. Whenever it reaches a valve, it adds that valve to the list and does not go past it. When nothing is left in the queue, the list holds the **boundary valves**.

You see: the queue, the visited nodes and the order of visits update live; found valves glow orange.

3 Close valves

All boundary valves are shut. The burst pipe is now cut off from the buildings behind those valves.

You see: the valves turn red with a small lock.

4 Check supply (2nd BFS)

BFS now starts at the **water tower**. It moves through open pipes only: it cannot pass a closed valve or the burst pipe. Every building it reaches is **supplied**; every building it cannot reach is **cut off**.

You see: supplied buildings in green (“WATER ACTIVE”), cut-off buildings in red (“SUPPLY CUT OFF”).

5 Report

A one-page summary: which pipe broke, which valves were closed, which buildings are affected, and the BFS visit order. Click **Download / Print PDF** to save it.

Shortcut

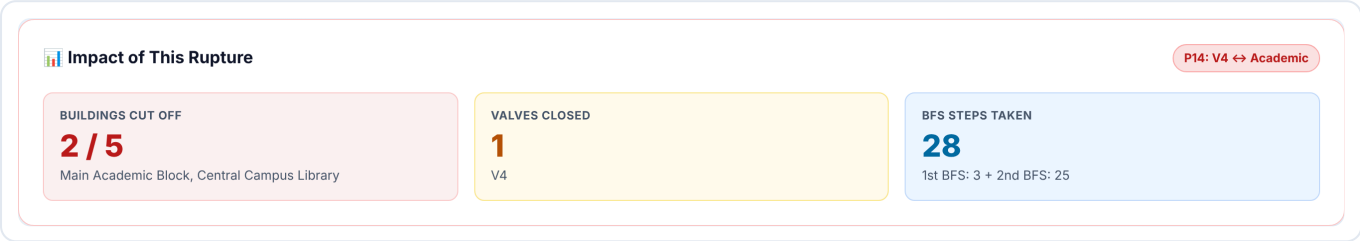
⚡ **Full Simulation** runs all 5 steps automatically. While a step is animating you can use **Play/Pause**, **Next**, **Prev** and **Reset** (keyboard: Space, →, ←, R).

5 A worked example: pipe P14 bursts

P14 is the short pipe from valve **V4** to the **Academic Block**.

P14 bursts → **End 1: V4 is a valve → close it** + **End 2: Academic Block has no other pipes → stop**

1. **1st BFS:** the two ends are V4 and the Academic Block. V4 is already a valve, so it goes straight onto the list. The Academic Block has no other pipes, so there is nowhere else to go. Result: **close V4 only**.
2. **Close valves:** V4 is shut.
3. **2nd BFS:** from the water tower, water reaches J1 → J2, J3, V1 … and every building *except* the two that sit behind V4.
4. **Result:** 2 of 5 buildings are cut off: the Academic Block *and* the Library. The Library loses water too, even though its own pipe is fine, because it shares valve V4.



The “Impact of This Rupture” panel that appears after every simulation.

The four ready-made scenarios on the Dashboard

Scenario	Pipe	Valves closed	Buildings cut off	Why
Water tower trunk (worst case)	P1	V1, V2, V3, V4	All 5	P1 is the only pipe leaving the tower.
Hostel service line (single building)	P3	V1	Hostel	The Hostel is a dead end behind V1.
Academic lateral (shared valve)	P14	V4	Academic, Library	Both buildings sit behind V4.
Bypass loop (looped pipe)	P16	V1, V2, V3, V4	All 5	See the box below.

Why does a burst on a loop pipe cut off everything?

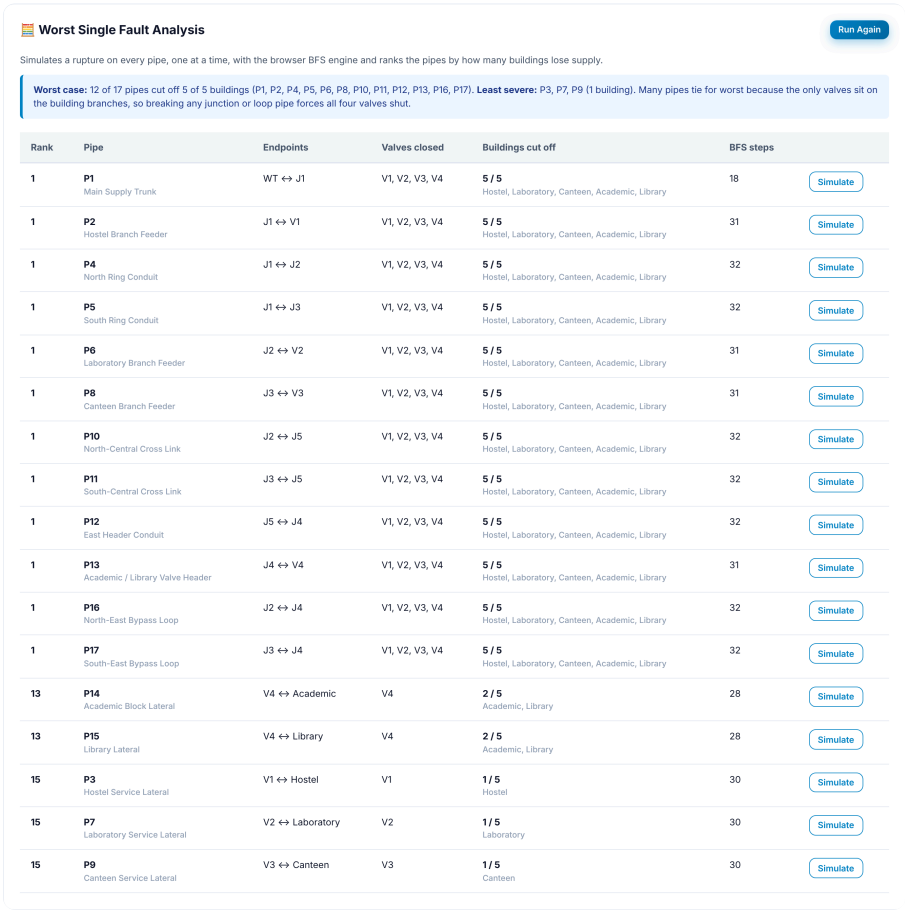
Water could still reach J4 another way (through J5), but the burst itself must be isolated, and the loop pipes have no valves of their own. Starting from J2 and J4, BFS spreads through every junction before it meets a valve, so the only way to shut off the leak is to close all four building valves. Adding valves on the loop pipes would allow a much smaller shut-off. This is a real lesson about where valves should be installed.

6 A quick tour of the website

Tab	What you can do there
Dashboard	Network summary, the four one-click scenarios, and the *Worst Single Fault* analysis.
Interactive Simulation	The map. Break a pipe, follow Steps 1–5, watch the queue and visited list, see the impact panel.
Algorithm Visualizer	A closer look at the BFS queue, the node being expanded, and a step-by-step event log.
BFS Algorithm Theory	The graph model, what BFS finds (and what it doesn't), and pseudocode for both BFS passes.
Isolation Report	The printable report for the last simulation, with the *Download / Print PDF* button.
About & Team	Course details, team roles and a short disclaimer.

Worst Single Fault

One button breaks each of the 17 pipes in turn and ranks them by how many buildings lose water. Each row has a *Simulate* button to replay that case on the map.



Worst Single Fault Analysis — Run Again

Simulates a rupture on every pipe, one at a time, with the browser BFS engine and ranks the pipes by how many buildings lose supply.

Worst case: 12 of 17 pipes cut off 5 of 5 buildings (P1, P2, P4, P5, P6, P8, P10, P11, P12, P13, P16, P17). **Least severe:** P3, P7, P9 (1 building). Many pipes tie for worst because the only valves sit on the building branches, so breaking any junction or loop pipe forces all four valves shut.

Rank	Pipe	Endpoints	Valves closed	Buildings cut off	BFS steps	
1	P1 Main Supply Trunk	WT ↔ J1	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	18	Simulate
1	P2 Hostel Branch Feeder	J1 ↔ V1	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	31	Simulate
1	P4 North Ring Conduit	J1 ↔ J2	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	32	Simulate
1	P5 South Ring Conduit	J1 ↔ J3	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	32	Simulate
1	P6 Laboratory Branch Feeder	J2 ↔ V2	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	31	Simulate
1	P8 Canteen Branch Feeder	J3 ↔ V3	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	31	Simulate
1	P10 North-Central Cross Link	J2 ↔ J5	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	32	Simulate
1	P11 South-Central Cross Link	J3 ↔ J5	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	32	Simulate
1	P12 East Header Conduit	J5 ↔ J4	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	32	Simulate
1	P13 Academic / Library Valve Header	J4 ↔ V4	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	31	Simulate
1	P16 North-East Bypass Loop	J2 ↔ J4	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	32	Simulate
1	P17 South-East Bypass Loop	J3 ↔ J4	V1, V2, V3, V4	5 / 5 Hostel, Laboratory, Canteen, Academic, Library	32	Simulate
13	P14 Academic Block Lateral	V4 ↔ Academic	V4	2 / 5 Academic, Library	28	Simulate
13	P15 Library Lateral	V4 ↔ Library	V4	2 / 5 Academic, Library	28	Simulate
15	P3 Hostel Service Lateral	V1 ↔ Hostel	V1	1 / 5 Hostel	30	Simulate
15	P7 Laboratory Service Lateral	V2 ↔ Laboratory	V2	1 / 5 Laboratory	30	Simulate
15	P9 Canteen Service Lateral	V3 ↔ Canteen	V3	1 / 5 Canteen	30	Simulate

12 pipes tie for worst (all 5 buildings), because the only valves are in front of buildings.

7 How the website is built

It is a plain website: **HTML** for the page, **CSS** for the look, and **JavaScript** for everything that moves. There is no framework and nothing to "build". An optional small **Python** server can run the same BFS on the server side.

File	What it does
`index.html`	The page: tabs, cards, buttons and text.
`css/style.css`, `css/animations.css`	Colours, layout, animations, phone layout and print layout.
`js/network-data.js`	The campus network: every node, every pipe, and the four scenarios.
`js/bfs.js`	**The BFS algorithm** (1st and 2nd pass) in JavaScript. It also records every step for the animation.
`js/simulation.js`	Runs Steps 1–5, Play/Pause/Next/Prev, and talks to the Python server when chosen.
`js/visualizer.js`	Draws the map (pipes, valves, buildings, colours).
`js/report.js`	Builds the Isolation Report.
`js/app.js`	Connects everything: tabs, buttons, panels, scenarios, worst-fault table.
`backend/app.py`	Small Python web server: serves the website and answers API requests.
`backend/bfs_engine.py`	The same BFS written in Python.
`requirements.txt`	List of extra Python packages. It is empty on purpose (only built-in Python is used); hosting sites expect the file to exist.

Two engines, same answer

DEFAULT

Browser engine

The BFS in `js/bfs.js` runs inside your browser. It needs no server and works even on a basic static host.

OPTIONAL

Python engine

Choose *Python REST API* under "Algorithm Engine". The browser sends the burst pipe to `backend/app.py`, which runs `bfs_engine.py` and sends the steps back. If the server can't be reached, the site says *"Server not reachable – using browser engine"* and carries on.

Both engines were checked on all 17 pipes and give identical results.

Run it on your computer

1. Install Python 3 (nothing else is needed).
2. Open a terminal in the project folder and run: `python backend/app.py`
3. Open `http://localhost:5000` in your browser. To open it on a phone, connect the phone to the same Wi-Fi and use the "Phone/Mobile" address the server prints.

8 Deploying on Render

Render (render.com) can host the website for free straight from GitHub. You can deploy it in one of two ways:

EASIEST

Option A: Static Site

- Hosts the HTML, CSS and JavaScript only.
- Uses the browser engine (works fully).
- Free and always awake.
- Python engine not available (shows the fallback message).

FULL VERSION

Option B: Web Service

- Runs `backend/app.py`: website *and* Python API.
- Both engines work.
- Free plan "sleeps" after about 15 minutes with no visitors; the next visit takes a little while to wake it.

Before you start

The code is on GitHub at **KrisJeffordImmanuvel/FAI_Sujith**. The latest version (with every change described here) is on the branch `claude/sweet-wozniak-yv6onk`. Either pick that branch in Render, or merge it into `main` first and pick `main`.

Option A: Static Site

1. Go to **render.com** and sign up / log in **with GitHub**.
2. In the Render Dashboard click **New +** → **Static Site**.
3. Choose the **FAI_Sujith** repository. If it isn't listed, use *Configure GitHub* / *Connect account* to give Render access to it.

4. Fill in the form:

Name	e.g. `kits-campus-water-bfs` (this becomes part of the web address)
Branch	`claude/sweet-wozniak-yv6onk` (or `main` after merging)
Root Directory	leave empty
Build Command	leave empty. If Render insists on one, type `echo "no build needed"`
Publish Directory	`.` (a single dot means "the main folder")

5. Click **Create Static Site** and wait until the status says **Live** (usually a minute or two).

6. Open the address Render shows, for example `https://kits-campus-water-bfs.onrender.com`.

Option B: Web Service (with the Python backend)

1. In the Render Dashboard click **New +** → **Web Service**, and choose the **FAI_Sujith** repository.

2. Fill in the form:

Name	e.g. `kits-campus-water-bfs`
Language / Runtime	**Python 3**
Branch	`claude/sweet-wozniak-yv6onk` (or `main` after merging)
Root Directory	leave empty
Build Command	`pip install -r requirements.txt`
Start Command	`python backend/app.py`
Instance Type	**Free**

3. **Do not** add a `PORT` setting. Render chooses the port itself (10000 by default) and passes it to the app, which reads it automatically.

4. Click **Create Web Service**. In the log you should see the banner *"Campus Water-Pipeline Fault Isolation Tool (BFS Engine)"*, then the status changes to **Live**.

5. Open the address. On the *Interactive Simulation* tab set **Algorithm Engine** to *Python REST API (backend server)*. On a laptop screen the badge at the top should say **"Python API Engine Online"**.

After deploying

- **Updates are automatic:** every time new code is pushed to the chosen branch, Render redeploys it (Auto-Deploy is on by default).
- **Quick check:** the Dashboard shows four scenario cards; "Load & Simulate" on the Hostel scenario should close only V1 and cut off only the Hostel; the report's *Download / Print PDF* opens the print dialog.

If something goes wrong

Problem	What to do
Web Service deploy fails with "no open ports detected"	Check the Start Command is exactly `python backend/app.py` and that the branch has the latest code (the app must read `PORT`).
Build fails: `requirements.txt` not found	The branch is out of date. Use the latest branch, or set the Build Command to `echo "no build needed"`.
Site shows an old version	Check which branch Render is using, then hard-refresh the page (Ctrl + Shift + R).
First visit is slow (Option B)	Normal on the free plan: the service was asleep and is waking up.
"Server not reachable – using browser engine" on a Static Site	Expected: a static site cannot run Python. Everything still works with the browser engine; use Option B if you need the Python engine.

Render's menus may be worded slightly differently over time; the field values above are what matter. The Python server uses Python's built-in web server, which is fine for a class demo but not meant for heavy traffic.

9 Good to know (limitations)

- **It is a connection model, not a hydraulic simulation.** It does not calculate water pressure or flow. "Supplied" simply means a building is still connected to the water tower through open pipes.
- **Nearest means fewest pipes, not shortest distance.** BFS ignores pipe length.
- **Valve placement decides the outcome.** With valves only in front of buildings, a burst on any junction or loop pipe cuts off the whole campus.
- **The water-loss figure in the report is a rough illustration**, not a measured value.